Hacktiv8

Predicting Student Academic Performance and Identifying Key Success Factors

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Dataset

The dataset is from two publicly available secondary school datasets, which are mathematics course performance and portuguese language course performance.

The dataset contains demographic, social, behavioral, and academic attributes of each student.

Problem

Educational institutions **struggle to identify** which students are likely to underperform before final exam results are known. Inefficient identification of “at-risk” students leads to higher failure rates.

Because academic outcomes are influenced by diverse factors, a predictive model is needed to understand how these factors influence final performance.

It will **help decision-making** and **early intervention** in schools and higher education institutions.

Objectives

1	**Preprocess**	Perform data preprocessing or feature engineering to prepare high-quality inputs for regression models.
2	**Implement**	Develop models using KNN, SVM, Decision Tree, Random Forest, and Boosting algorithms with pipeline.
3	**Analyze**	To analyze and identify how each factors influence students' final academic performance.
4	**Predict**	Estimate students' final grades from various factors.

Final Model: Gradient Boosting Regressor

Best hyperparameters:

- learning rate: 0.05
- max_depth: 3
- min_samples_leaf: 5
- n_estimators: 100
- subsample: 0.8

→ High predictive accuracy as the model has R2 of 0.82 and MAE of 0.91.

→ Robust to non-linearity because Gradient Boosting captures complex relationships between student's features with the final grade.

→ Unbiased predictions as the residual distribution is tightly centered around 0, indicating that the model does not over-predict or under-predict for the general student population.

Hugging Face

1

Exploratory Data Analysis

2

Prediction